

Quantile XGBoost for NYC Construction Scheduling Risk

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Abstract

Predicting construction durations across a diverse portfolio such as DOB NOW: Build Job Filings; 2016–2026; $N = 911,251$ requires moving beyond standard average estimates (Section I). In Section II, I work towards a practical framework for quantifying scheduling uncertainty using a filtered modeling sample ($n_{model} = 221,519$). After baselining a standard regression XGBoost model (Section III.I), I test a QXGBoost approach [Yin et al., 2023] (Section III.II), configured for the large variance and high number of features present in the data. As detailed in Section III, the models identified Neighborhood Tabulation Area (NTA) as the strongest predictor, followed by initial cost and construction floor area (consistent with industry literature). On a chronological test set (July 2024–May 2026; $n_{test} = 44,304$), the model achieved 84.4% bracket coverage, with the worst-case upper bound capturing true durations 93.4% of the time. With a fully reproducible code appendix (Section VI), the current model/analysis may serve as a useful descriptive tool while laying some of the groundwork for a deployable, production-ready scheduling application (Section IV).

I Introduction

Background & Literature

Construction informatics considers risk management and uncertainty quantification as important attributes of project success [Turk, 2006]. Knowing not just whether a project will finish, but when, and within what range, is among the most consequential problems a general contractor (GC), owner, or other stakeholder faces. Assigning a single number to project duration can, in many realistic contexts, be problematic. Cost overruns, contract penalties, financing issues, and subcontractor sequencing all depend on timeline reliability. A probabilistic bracket is more honest, and therefore useful, than a point estimate, a view advocated for in recent literature [Nguyen et al., 2013].

Historical approaches to construction duration forecasting used linear trendlines segmented by building type [Martin et al., 2006] or early artificial neural networks (ANNs) in limited geographic contexts [Bhokha and Ogunlana, 1999]. These methods achieved reasonable accuracy with clean samples: road/highway-only datasets [Peško et al., 2017, Qiao et al., 2019], single-city residential portfolios [Allah et al., 2015], or high-rise towers only [Nguyen et al., 2013]. Homogeneous samples give models more of a “fighting chance” [Trijeti et al., 2023]. When the range of inputs is narrow, the distribution of duration is easier to learn, though some use-cases may not match those assumptions.

This project analyzes the NYC Department of Buildings DOB NOW: Build – Job Application Filings [New York City Department of Buildings, 2026], which as of June 2026 contains approximately 911,251 job filings spanning residential renovations, commercial fit-outs, structural alterations, mechanical upgrades, and more. Construction machine learning literature has matured considerably alongside data availability. Gradient boosting approaches can outperform ANNs on prediction tasks [Zhang and Li, 2024, Chen and Guestrin, 2016], while large-scale hybrid approaches integrating textual and numerical features have achieved high prediction accuracy [Liu and Luo, 2026]. The theoretical foundation for moving to quantile estimation is well established [Koenker and Bassett, 1978], and recent implementations of QXGBoost have shown promise for uncertainty quantification in complex settings [Yin et al., 2023].

Limitations

This analysis is bounded by significant assumptions and limitations that should be stated plainly at the start (following the modeling discipline advocated for by Thompson [2022]; Hoerl and Redman [2023]; Hoerl and Redman [2024]). First, the timeframe for this project made some choices prohibitive despite their theoretical advantages for right-censored time-to-event data [See Qiao et al., 2019, for other approaches]. Second, this analysis suffers inherent survivorship bias. I filter to projects with a recorded sign-off date, excluding stalled or abandoned works in a way that a survival framework would handle through censoring. Third, data quality issues persist throughout. Initial project costs are routinely low-balled to minimize DOB fees, it’s clear in the data, and consistent with historical patterns of cost bias construction forecasting literature [Teicholz, 1993]. Fourth, while “native” QXGBoost is utilized here (`objective="reg:quantileerror"`), recent advances recommend arctan pinball loss [Sluijterman et al., 2024] or Huber-smoothed loss functions [Yin et al., 2023] (I barely understand the distinction/math, and wouldn’t know how to go about implementing it in R, just thought it should be mentioned). Lastly, consider my limited knowledge of statistics, machine learning, and the NYC construction domain. Though I try to be thorough, this project will inevitably have errors, and I welcome (highly encourage) criticism and honest

feedback.

In its current form, the model is best understood as a proof of concept, that could be extended to a production-ready scheduling application. The objective is not to replace expert judgment but to provide a grounded descriptive tool that a contractor could use as a starting point [Hoerl and Redman, 2024, Thompson, 2022].

II Data & Methods

II.I The DOB NOW Dataset and Duration Definition

The DOB NOW system (transition from the BIS platform) is a permit/job management platform through which recent construction jobs in the five boroughs flow. The full dataset, available through NYC OpenData, is updated daily and as of June 2026 contains over 900,000 jobs across 95 columns [New York City Department of Buildings, 2026]. Ten columns including `job_description`, `owner_type`, `green_roof`, and `solar_roof` were added (merged from elsewhere I believe) on May 27, 2026, expanding the available feature space meaningfully for this analysis.

Duration definition

Following the DOB’s guidance on project lifecycles, and exercising some judgement [New York City Department of Buildings, 2024], `duration` is defined here as the number of days from first permit issuance to sign-off date. The first permit date marks the moment construction really commences; the sign-off date marks completion and regulatory closure. This definition is clean and business-relevant. The lag from filing to first permit, `filing_to_permit` captures the procurement phase, and is therefore placed on the feature side of the model, not the target side. It carries real signal about mobilization and complexity.

Why this sample is difficult

Compared to other literature that analyzes more homogeneous samples such as: road/highway construction only [Peško et al., 2017, Qiao et al., 2019], high-rises only [Nguyen et al., 2013], or projects split by building type [Trijeti et al., 2023], the full DOB NOW dataset spans a variety of scopes (not LAAs, not electrical, not elevator). A sidewalk shed permit, a gut renovation of a Midtown skyscraper, a mechanical upgrade in the Bronx, and a foundation project in Brooklyn all live in the same dataset. This heterogeneity is realistic but difficult for modeling, and more trimming/thoughtful cleaning would’ve been better. Figure 1 shows the challenging distribution.

The mean duration in the modeling dataset is 334 days, the median 249 days, and the standard deviation 289.73 days (Table 2). The gap between mean and median reflects the right skew that characterizes this broad sample/context, and as a result, makes average-based prediction misleading as a planning tool.

II.II Data Pipeline & Filtering

Collapse to initial filings

Each job can generate multiple filings. The `job_filing_number` encodes filing type with a suffix letter: “I” for Initial, and various letters for Post-Approval Amendments (PAA) and Subsequent filings. The unit of analysis is one Initial Filing per job ID. PAAs are excluded because they

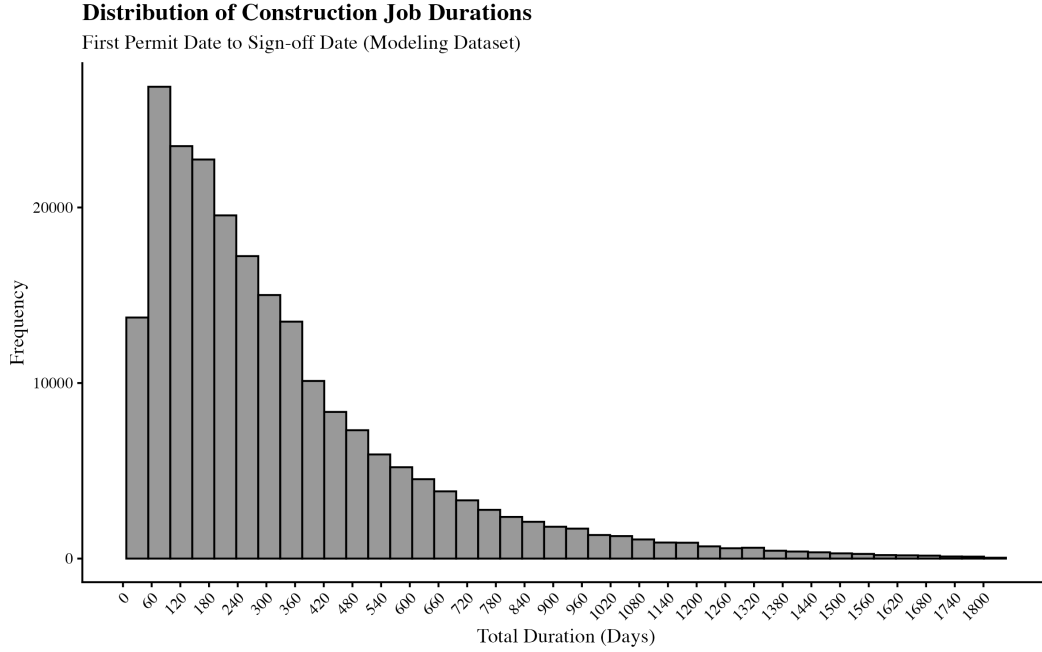


Figure 1: Distribution of Construction Job Durations.

reflect amendments that occur after the project begins, and they are not observable at the time of scheduling. Subsequent filings submitted before the first permit are retained and counted as a feature (`subsequents_at_permit_time`). This collapse removed 368,078 rows from the raw dataset.

Filtering to completed jobs

Jobs are retained only when a first permit exists (the clock has started in the field), a sign-off date exists (construction is completed/closeout), `duration` is positive, and `duration` falls between 30 and 1,825 days. Below 30 days is likely a data integrity error or a trivially minor job where duration risk is less meaningful anyway. Above five years, a single permit on a non-skyscraper project is almost certainly a stalled, restarted, or otherwise anomalous job. Large commercial projects in NYC tend to operate under multiple permit filings (the new JP Morgan tower at 270 Park, for example, carries hundreds of associated filings, see [New York City Department of Buildings \[2026\]](#)). This survivorship filter is the main source of the bias acknowledged. A Random Survival Forest could be a better approach for right-censored time-to-event data, or something similar (See [Qiao et al. \[2019\]](#)). I did not pursue this because of limitations related to time, knowledge, and computing power on my M1 MacBook Air.

Substantive Work Filtering

Sidewalk shed, scaffold, and fence permits with no substantive underlying work are excluded. Negligent owners routinely leave these structures standing indefinitely, and they contribute only noise to duration modeling. The resulting modeling dataset contains $n_{model} = 221,519$ jobs with approximately 70 features. More trimming could yield better results.

II.III Feature Engineering

Cost

`initial_cost` is self-reported and widely underestimated to reduce DOB fees. Values below \$1,000 are treated as implausible and set to missing. The remainder is winsorized at the 95th percentile and log-transformed (`log_initial_cost`). A `cost-per-square-foot` flag was constructed to identify underreporters, though models made little use of it.

Scale Total construction floor area is similarly log-transformed (`log_floor_area`) after replacing zeros with missing and winsorizing (consider facade work or other substantive, low-footprint work and handle this better). Floor area and initial cost has been consistently identified as among the strongest predictors of construction duration in similar literature [Jarkas, 2016, Zhang and Li, 2024], they rank highly in importance here as well.

Inspection counts

`special_inspection_count` and `progress_inspection_count` are counts of the comma-separated inspection requirement fields. High-hazard inspection types (structural, concrete, foundation, excavation, demolition, etc.) are flagged separately as `high_hazard_special_count` and `high_hazard_progress_count`.

Text analysis

A basic (understatement) keyword search on the `job_description` field flags high-friction regulatory and physical complexity signals: asbestos, facade/Local Law 11, landmark designation, gut renovation, structural work, elevator, sub-grade, etc. and scope-related verbs like “erect” or “vertical enlargement” (high friction) vs. “repair in kind” or “replace in kind” (low friction). This approach is similar to Liu and Luo [2026], who found hybrid NLP-ML models outperformed purely numerical approaches on large-scale construction task prediction.

Geographic features

`borough`, `nta` (Neighborhood Tabulation Area), and `council_district` are encoded as factors. Borough is the coarsest level: Manhattan’s density, regulatory complexity, and landmark building stock make it objectively harder to build in than other boroughs. NTA provides meaningful sub-borough granularity; Midtown-Times Square operates under very different constraints than Flushing or Bay Ridge. NTA emerged as the models’ single strongest predictor.

Time features

Permit month, permit year, a winter-start flag (November–February), and a COVID-period flag (January 22–August 8, 2020, bracketing the construction pause of March 22–June 8, 2020) are included.

Complexity score

An additive composite of structural, foundation, mechanical, earth-work, demolition, and excavation work-type flags plus high-hazard inspection counts and scope signals. This is not a substitute for the individual features but gives the trees an additional feature to split on.

Work type flags

Binary indicators for all substantive work types (general construction, mechanical systems, structural, plumbing, sprinkler, boiler, earth work, foundation, protection/mechanical methods, support of excavation, etc.) are included in full. Table 1 shows *selected* descriptive statistics from the modeling dataset. Full feature list is in Section VI.

II.IV Why Quantile Regression, and Why XGBoost

The problem with average estimates

Standard regression minimizes squared error and produces a conditional mean. For a symmetric, low-variance distribution, the mean is a reasonable central estimate. For a right-skewed distribution with a standard deviation nearly as large as the mean (as is the case here), that is difficult. Also, a GC bidding on a job does not need to know the expected value of duration averaged across all possible states; they’d need to know: what is a realistic worst case? What is a reasonable best case? A risk interval provides both. Assigning a single number to a project duration is dangerous; a bracket is better suited to the context, and the ability to flag jobs that are likely to run long is more actionable than an inaccurate average prediction [Nguyen et al., 2013].

Quantile Regression

Koenker and Bassett [1978] introduced quantile regression as an alternative to OLS for different error distributions. Rather than minimizing a squared loss, quantile regression minimizes “pinball loss” to estimate conditional quantiles. For right-skewed NYC construction durations, spanning various scopes/work types, this is the theoretically appropriate target: the model is not trying to predict the average duration conditioned on features, but rather the P05 (best case) and P85 (worst case) thresholds, yielding an 80% risk interval.

eXtreme Gradient Boosting

Chen and Guestrin [2016] introduced XGBoost as tree boosting system that consistently achieves state-of-the-art results on tabular data. For a large dataset ($n_{train} = 177,215$) with many features, I considered XGBoost a good place to start. It handles mixed data types, missing values, and scales efficiently. Gradient boosting methods can outperform ANNs, SVMs, and k-NN on construction duration prediction while being orders of magnitude faster than ANN [Zhang and Li, 2024]. The QXGBoost model for this project is adapted from Brownlee’s approach and R implementation from Hoerl [2026].

RF-emulation architecture

The XGBoost and QXGBoost architectures here were configured to emulate a random forest through aggressive column and row subsampling. The motivation: without subsampling and feature restriction, NTA, initial cost, and floor area would dominate every tree (they did anyway). By restricting the features available at each split (`colsample_bytree = 0.6`), the model is forced to find duration signal in less dominant features work types, inspection counts, text flags, time variables, even if that signal is weaker. Or at least that was my intention. Whether this materially improved coverage is debatable, but it produced a feature importance profile that reveals cross-quantile differences consistent with domain intuition (see Figure 3), and it remains insightful.

Survival analysis as a theoretical better approach

A Random Survival Forest or parametric time-to-event model is probably the correct approach for

Table 1: Selection of Descriptive Statistics of the NYC Construction Modeling Dataset ($n_{model} = 221,519$). Initial Filings; $n_{features} \approx 70$, subset shown below, final available in Section VI; Dates range from 9/20/2016 to 5/1/2026.

Variable	Statistic / Category	Value / Count
Target Variable	Mean Median	$\bar{x} : 334 \quad \tilde{x} : 249$
Duration (Days)	Min – Max	30 – 1825
	Standard Deviation	$s : 289.73$
Log Duration	Mean Median	$\bar{x} : 5.457 \quad \tilde{x} : 5.517$
	Min – Max	3.401 – 7.509
	Standard Deviation	$s : 0.879$
Continuous Features	Mean Median	Min – Max (NAs)
Log Initial Cost	$\bar{x} : 10.338 \quad \tilde{x} : 10.427$	6.909 – 13.162 (11,408)
Log Floor Area	$\bar{x} : 6.773 \quad \tilde{x} : 7.091$	0.010 – 10.600 (22,865)
Progress Inspection Count	$\bar{x} : 2.055 \quad \tilde{x} : 1.000$	0 – 24 (0)
Special Inspection Count	$\bar{x} : 1.379 \quad \tilde{x} : 1.000$	0 – 19 (0)
Discrete Features	Top Categories	Count
Borough	Manhattan	100,437
	Brooklyn	51,395
	Queens	41,615
	Bronx	16,713
	Staten Island	11,359
Council District	04	31,178
	03	19,192
	01	13,598
	Other (Combined)	136,980
NTA	Midtown-Times Square	18,539
	Midtown South-Flatiron	7,506
	Upper East Side-Carnegie Hill	7,245
	East Midtown-Turtle Bay	6,697
	Other (Combined)	174,989
Review Building Code	2014	95,147
	2022	68,617
	1968	38,592
	Prior to 1968	18,235
Owner Type	Corporation	114,265
	Individual	54,337
	Partnership	27,149
	Condo / Co-Op	12,037
	Non-Profit	8,200
Filing Review Type	Professional Certification	141,448
	Standard Plan Examination	80,071

Source: NYC OpenData & NYC Department of Buildings, DOB NOW: Build Job Application Filings [New York City Department of Buildings, 2026].

construction duration data [Qiao et al., 2019]. Survival analysis handles right-censored observations (projects that are ongoing and have not yet signed off), rather than discarding them as I did. The QXGBoost approach does trade theoretical correctness for practical explainability: risk intervals in the quantile model are interpretable as direct probability statements, while survival functions require additional translation for a non-technical stakeholder (I don’t fully understand it either). For a descriptive, preliminary analysis, I’m satisfied with QXGBoost.

II.V Post-Hoc Calibration

Following the recommendations of Romano et al. [2019] on Conformalized Quantile Regression and the empirical findings of Yin et al. [2023] regarding gradient boosting coverage, a “post-hoc calibration” was performed. To achieve the target 80% overall bracket coverage on the hold-out test set, the nominal training quantiles were shifted from the symmetric P10/P90 levels to a P05/P85 bracket.

This may seem like “shifting the goal posts”, but my understanding is that post-hoc calibration is statistically valid. This adjustment is substantiated by Pouplin et al. [2024], who argue that intervals constrained to be symmetric around the median are worse for realistic skewed distributions. Since NYC construction duration data is heavily right-skewed (many small jobs, a long tail of multi-year jobs), a symmetric interval imposed a distribution assumption that didn’t capture the true risk profile. The shift to P05/P85 prioritizes an ambitious “best case” lower bound (P05) while providing a safer, more conservative “worst case” upper bound (P85) that better captures the long tail of construction delays.

II.VI Model Specification

Train/test split

A chronological 80/20 split was used: the first 80% of jobs by first permit date form the training set ($n_{train} = 177,215$; September 2016 to July 2024), the remaining 20% the test set. This reflects how the model would actually be used in practice: trained on historical projects, applied to future ones. The test set covers July 2024 through May 2026 ($n_{test} = 44,304$).

Table 2 shows parameter specifications for the Quantile XGBoost models ($\alpha = .05$, $\alpha = .50$, and $\alpha = .85$).

The standard XGBoost model was specified a bit different (See Section VI). Both models were trained on log-transformed duration; predictions were exponentiated back to days for evaluation. All categorical variables (`nta`, `borough`, `council_district`, `job_type`, `building_type`, `owner_type`, `filing_review_type`, `review_building_code`, etc.) were passed as native XGBoost factor types.

Table 2: Hyperparameter Specs for the QXGBoost Models

Hyperparameter	R Argument	Value
Booster Type	booster	"gbtree"
Objective Function	objective	"reg:quantileerror"
Learning Rate	eta	0.05
Maximum Depth	max_depth	4
L2 Regularization	lambda	10
Subsample Ratio	subsample	0.8
Column Subsample	colsample_bytree	0.6
Maximum Rounds	nrounds	1500
Early Stopping	early_stopping_rounds	100

Note: The `quantile_alpha` parameter was modified across training runs ($\alpha = 0.05$ and $\alpha = 0.85$) to generate the asymmetric intervals for context-specific scheduling scenarios.

III Results

III.I Standard XGBoost Regression

The Standard XGBoost model, minimizing mean squared error on log-transformed duration, establishes a base performance level (a modest one) on this dataset. Table 3 presents the chronological test set evaluation results.

Table 3: Standard XGBoost Evaluation ($n_{test} = 44,304$)

Evaluation Metric	Achieved Value
Error Metrics	
R-Squared (R^2)	0.087
Root Mean Squared Error (RMSE)	119.9 Days
Mean Absolute Error (MAE)	93.0 Days
Prediction Accuracy Thresholds	
Error < 30 Days	20.1%
Error < 60 Days	40.4%
Error < 90 Days	58.4%
Error < 180 Days	88.2%

An R^2 of 0.087 might appear poor, but consider it in context. The model explains less than 9% of the variance in test-set, though this result is not so surprising with a broad portfolio. For comparison, [Liu and Luo \[2026\]](#) report $R^2 = .329$ for their k-NN benchmark; [Zhang and Li \[2024\]](#) demonstrated that baseline models for specific building components, such as main towers and corridors, can result in negative R^2 values. Stop-work orders, material delivery disruptions, labor disputes, weather events, and regulatory complications, all major drivers of actual duration, are not in the available feature set. For future analysis, better operational data and sub-samples should be evaluated.

The accuracy threshold distribution is more instructive, and this interpretation inspired the quantile

extension. The model predicts within 90 days for 58.4% of jobs and within 180 days for 88.2%, performance that, while modest, could have descriptive value for broad portfolio planning purposes. An MAE of 93 days on a distribution with a standard deviation of 289 days represents meaningful signal above predicting the mean for every job. Feature importance is similar to the quantile models.

III.II Quantile XGBoost

The quantile approach reframes the task from point prediction to coverage: the goal is not to minimize average error but to produce an 80% risk interval; a [05, P85] bracket that contains the true duration for 80% of future out-of-sample jobs. Table 4 presents the coverage results; note that the quantiles were shifted from a symmetric distribution assumption to achieve the target coverage.

Table 4: Quantile Gradient Boosting Evaluation: Out-of-Sample Coverage ($n_{test} = 44,304$)

Metric	Target Coverage	Achieved Coverage
Percent of jobs slower than Best Case (P05)	90.0%	91.0%
Percent of jobs faster than Worst Case (P85)	90.0%	93.4%
Overall Bracket Coverage (P05–P85)	80.0%	84.4%

The model achieves 84.4% overall bracket coverage against a target of 80%, exceeding the target on both bounds individually. The upper-bound (worst-case) performance is even stronger. 93.4% of test jobs completed in fewer days than the P85 prediction, meaning the worst-case bound is conservative and therefore useful for risk management, a GC planning for the worst case will be late in fewer than 7% of scenarios. This is the quality of quantile regression that makes the risk interval useful; it is more important to cover the right tail (catastrophic delays) than the left (surprisingly fast completions), and the calibrated P05/P85 bracket help achieve this. Figure 2 visualizes the risk intervals on a random sample of 100 test-set jobs.

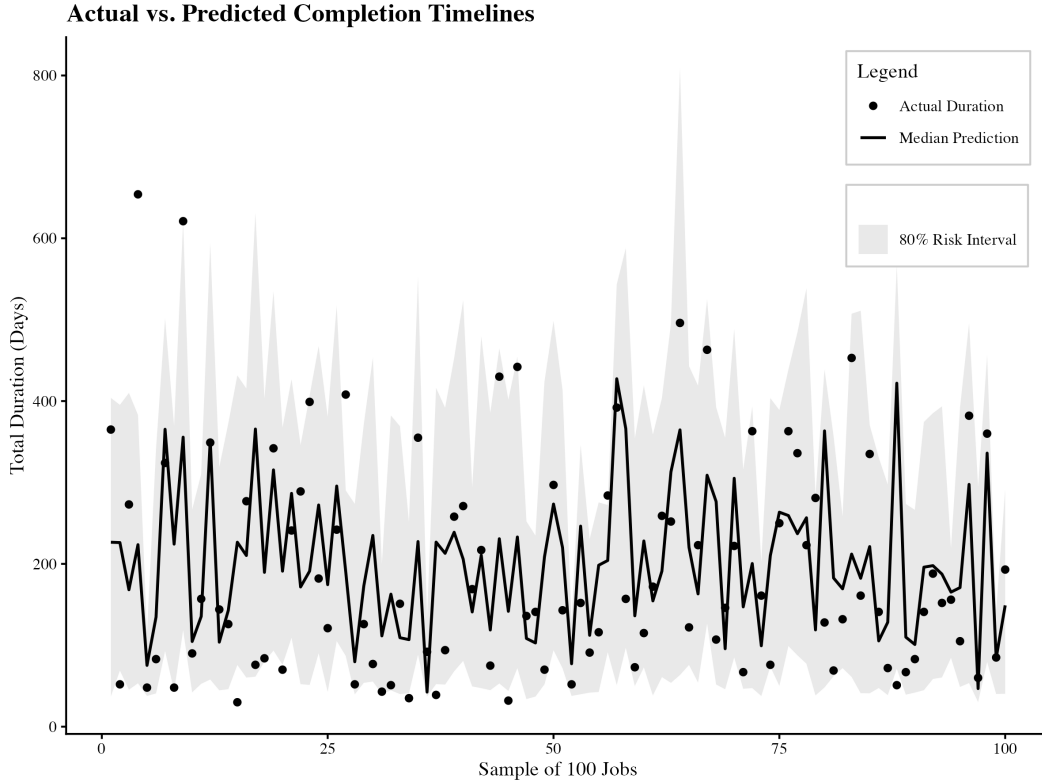


Figure 2: Risk Intervals.

The ribbon plot shows several important characteristics of the model. First, the intervals are wide, which is an honest reflection of the dataset, and my modeling abilities. In my defense, a narrow interval in this context could be overconfident and wrong. Second, the median (P50) prediction tracks actual durations reasonably well, though some of the longest-duration jobs are still underpredicted (quantile calibration partially addresses this). Third, the outliers above the worst-case bound tend to be the longest-duration jobs, precisely the projects where uncertainty is highest and where a survival framework that models censoring explicitly could be valuable. Figure 3 displays the QXGBoost feature importance for the P05, P50, and P85 models.

Several patterns are worth noting. NTA is the single strongest predictor in every model and across quantiles. Its importance is largest in the worst-case (P85) model. This aligns with industry literature establishing that project location is a primary driver of construction timelines [Jarkas, 2016, Zhang and Li, 2024]. In NYC specifically, this finding captures the compounding regulatory, logistical, and density-related constraints that vary dramatically across neighborhoods. Midtown-Times Square is fundamentally harder to build in than Flatbush, and the model has learned this from ~170,000 training jobs.

Cost and floor area

`log_initial_cost` and `log_floor_area` are the second and third tier of predictors, consistent with Jarkas [2016] and Zhang and Li [2024]. Notably, `log_floor_area` has slightly higher relative importance in the median and worst-case models than in the best-case model, while `log_initial_cost` is more prominent in the best-case model. This suggests that project scale is a better predictor of central and long-tail outcomes, while reported cost (problematic as it is in this data) is more informative for the left tail of fast completions.

Feature Importance Distribution across Risk Quantiles

Comparing Gain P10, P50, and P90 models

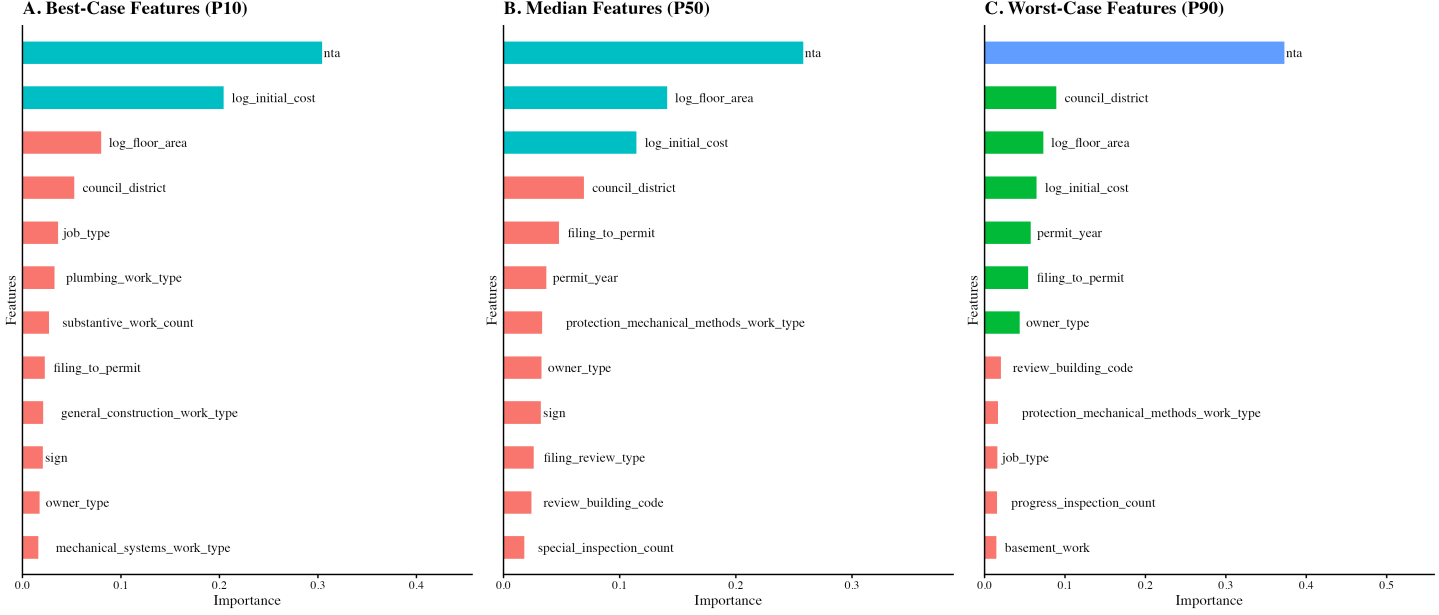


Figure 3: QGB Feature Importance.

`filing_to_permit` is a median and worst-case signal. The lag from filing to first permit issuance appears more prominently in the P50 and P85 models than in the P05 model. Slow approval to mobilization is a friction signal that predicts longer projects more than it predicts faster ones.

Work-type flags carry more signal in the worst-case model. `plumbing_work_type`, `protection_mechanical_methods_work_type`, and `basement_work` rank higher in the P85 model than in the P05 model, suggesting that complexity features are better at predicting delays than at predicting rapid completions. A job involving plumbing or below-grade work is not necessarily fast, but it is reliably at risk of being slow. This is a substantive finding with practical implications for risk flagging.

`permit_year` and `review_building_code` both appear in the upper quantile models, reflecting the temporal, regulatory, and building-code structure of the data. I used the `Ckmeans.1d.dp` package in order to force three importance plots together, which appears to have done some clustering. Though I did not pursue this further, it should be.

IV Summary & Final Thoughts

This analysis is more of a descriptive and preliminary approach for quantifying NYC construction scheduling risk using QXGBoost on DOB NOW Job Application Filings. The important findings are: (1) a data pipeline that collapses nearly one million filings into a somewhat cleaner modeling sample of 221,519 signed-off jobs, (2) a feature engineering approach that combines geographic granularity, text-extracted regulatory signals, continuous scale measures, and work-type complexity indicators, (3) a QXGBoost implementation with post-hoc calibration that achieves 84.4% bracket coverage on a chronological holdout, and (4) a cross-quantile feature importance analysis that reveals asymmetries in what predicts best-case vs. worst-case outcomes.

Avenues for further research

An natural extension is survival analysis. A Random Survival Forest, perhaps using `ranger` [Wright and Ziegler, 2017] or a hazard model [Qiao et al., 2019] would handle the survivorship bias inherent in filtering to completed jobs. Currently, the analysis discards all ongoing projects, or projects that weren’t finished. In a production application, those projects are exactly the ones a stakeholder is trying to forecast. Incorporating censored observations with survival methods would improve both the validity and the scope of inference. Beyond survivorship, the model would benefit from a cleaner sample. Splitting the analysis by job type (hard because most jobs are Alterations) or by borough would give a sub-model a better signal-to-noise ratio [Trijeti et al., 2023]. A cleaner, reliable cost variable, could reduce the measurement noise from underreporting. The text analysis on `job_description` should be extended significantly, following a more sophisticated NLP approach similar to Liu and Luo [2026]. Finally, implementing arctan pinball loss [Sluijterman et al., 2024] or a Huber-smoothed quantile objective [Yin et al., 2023] could improve performance (again, not my area of expertise).

Ethical considerations

In the spirit of responsible analytics articulated in Hoerl and Redman [2024]; Hoerl and Redman [2023]; and Thompson [2022], any deployment of a predictive scheduling model needs safeguards in place. First, a contractor or industry expert’s judgment should always come before a model’s output for any specific job. The model has learned from broad patterns. The expert knows the specific site conditions, crew, subcontractor relationships, and current regulatory climate. Models can be used as a guide for decision-making and resource allocation, but never as a substitute for expert judgment. Also related, see “social dimension” & “social aspects” of construction informatics in Turk [2006].

Second, but most important, construction safety must remain the absolute priority in the field. A predictive model should never be used to increase pace at the expense of safe working conditions. If a project is running behind the model’s “best case” bound, the correct response is better management, not cutting corners on safety. The objective of Machine Learning for construction schedules is to provide a reference that supports the human element of construction, not one that compromises it. A model output that says a job should be done in 200 days is not a mandate; it is a piece of information among many that a thoughtful professional weighs alongside everything else they know.

V References

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